

MICS Plus Tabulation

A step-by-step guide to using the application

MICS Team

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Start here

This guide helps you turn the files supplied by your survey team into Excel tables. You do not need to write calculations or change computer code. Follow one step at a time. Ask your survey lead for help if a file or a result is different from what you expect.

Your route through the application

1. **Data Preparation** - choose the files for your survey.
2. **Tabulator** - calculate one table.
3. **Consistency Checks** - check that its totals agree.
4. **Write to Excel** - save the reviewed table.

The later pages explain running several tables, downloading data, and optional viewing tools. You can leave those until you have completed your first table.

Words used in this guide

- **Click:** press the left mouse button once. On a touchpad, tap once.
- **Tab:** a section name across the top of the application.
- **Browse:** open a window to find a file on your computer.
- **Folder:** a place that holds files. A preparation folder may contain smaller folders.
- **Sheet:** one page of an Excel workbook, such as 1.1a.
- **Wave:** one round of the survey. Use the wave your team gave you.

About the pictures. These are actual application screenshots from version 0.2.7, using Mongolia Wave 1 as an example. Your file names and results will differ. Use the files for your own survey. Screenshots show a temporary example output folder; your project normally saves to **micsPlusTableR-output**. In the HTML guide, click a picture to enlarge it; press **Esc** to return. In the PDF, use your reader's zoom control if small labels are hard to read.

Looking for package functions? Use the [package reference manual](#) and [package documentation](#). Those documents cover programming and technical details.

1. Open the application

The first time: ask a support person to help

Your computer needs **R**, **RStudio**, and a web browser. Ask your support person to install them and give you a survey project folder. You need internet access to install the application package from GitHub.

1. Open your survey project folder in **File Explorer** on Windows or **Finder** on a Mac. Double-click its file ending in **.Rproj** to open RStudio.
2. Find the panel named **Console**. Click beside the **>** prompt.
3. Copy the first line below into the Console and press **Enter** (**Return** on a Mac). Wait for the **>** prompt to return. Then run the second line and wait again:

```
install.packages("remotes")
remotes::install_github("barisr/micsPlusTableR")
```

4. If installation asks a question or reports an error, ask your support person to help before continuing.
5. When installation finishes, click **Session**, then **Restart R** at the top of RStudio.

Each time you work

1. Open your survey's **.Rproj** file as above.
2. In the **Console**, paste this line and press **Enter**:

```
micsPlusTableR::run_app()
```

3. The application opens in a browser. Click **Data Preparation** at the top if it is not already selected. The next page lists the files you will need.

You are ready when: you can see **Data Preparation**, **Tabulator**, **Consistency Checks**, and **Write to Excel** across the top. Click **User's Guide** to read these instructions inside the application. Use **Open full guide** there if you want a separate browser tab.

While working: keep RStudio and the application open. A busy Console without a **>** prompt is normal while the app runs. Do not paste more commands there. Do not refresh the browser; you may lose work not yet written to Excel.

No browser window? Find the address beginning **http://127.0.0.1:** in the Console. Copy the complete address into your browser's address bar and press **Enter**. The number at the end can change each time you start.

Install again only when your survey lead asks you to update. For installation help, show your support person the Console message and the [package setup instructions](#).

2. Gather your survey files

Ask your survey lead for these **four items for the same country, period, and wave**. Keep them together in a folder you can find again.

Item to obtain	How to recognize it	Where it goes in the app
Household data	Name often ends in hh.sav	Select household file
Household members data	Name often ends in hl.sav	Select household members file
Tabulation plan	Excel file ending in .xlsx	Select tabulation file
Preparation folder	Folder containing the team's preparation file	Select preparation folder

Your team may supply longer names. Do not rename the files just to match the pictures. You do not need to open the **.sav** files or the preparation code.

How to choose a file

1. Click **Browse...** beside the relevant item in the application.
2. In the window that opens, find the folder your team gave you. If it was downloaded, first look in **Downloads**.
3. Double-click a folder to open it. Click the required file once to select it.
4. Click **Open**. Wait until its name appears in the application and the upload finishes. Repeat for the other files.

How to choose the preparation folder

1. If the team supplied a ZIP file, extract it first: on Windows, right-click it and choose **Extract All**; on a Mac, double-click it.
2. Beside **Select preparation folder**, click **Browse...**
3. Select the folder that contains the preparation file, then click **Upload**, **Select**, or **Open**, depending on your browser. Select the folder itself, rather than an individual file inside it.
4. If the browser asks to upload the folder's files, check its name and confirm.

Check before continuing: use the complete preparation folder supplied by your team. Keep any folders inside it, such as **FIES-inputs**, in place. If you cannot tell which files belong together, ask your survey lead.

3. Choose the files and click Set

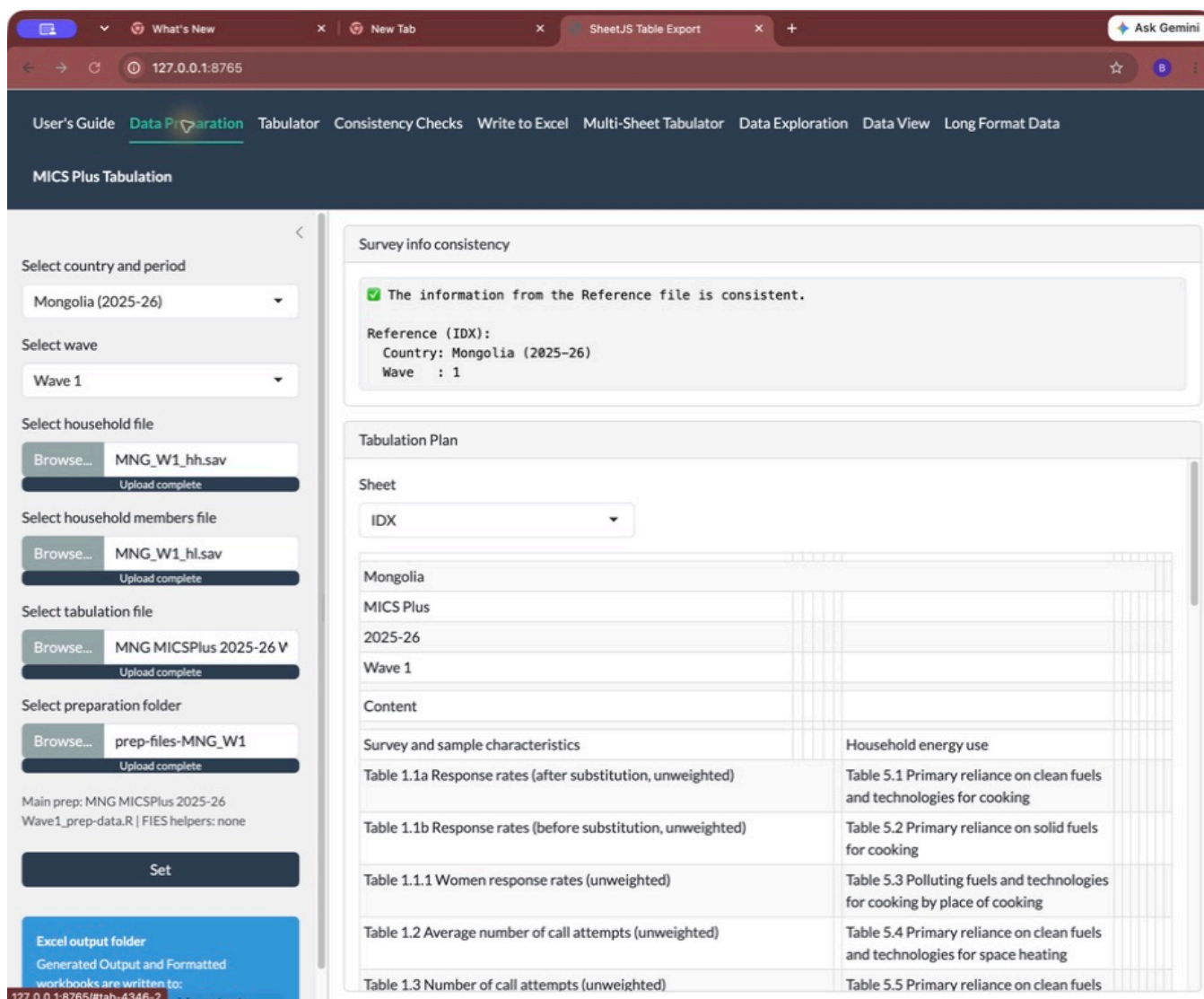


Figure 1: Data Preparation: survey choices and four Browse controls are on the left; the survey match message and Excel plan are on the right.

1. Click **Data Preparation**. Under **Select country and period**, click the small arrow and choose your survey. Choose the matching **wave** below it.
2. Use the four **Browse...** controls as described on the previous page.
3. Read **Main prep** below the folder control. Check that it names your team's preparation file. **FIES helpers: none** is normal for surveys without FIES.
4. Click **Set** once. If it is below the visible area, scroll down inside the left panel. Wait for preparation to finish.
5. Read **Survey info consistency** at the upper right. Check the country, period, and wave. The example says the reference information is consistent.

Ready to continue: the survey details agree and there is no preparation error. A matching survey message alone does not resolve a separate error. If a mismatch or error appears, check your choices or ask your survey lead.

4. Calculate and look at one table

The screenshot shows the MICS Plus Tabulation web application. On the left, the 'Select a sheet' dropdown is set to '1.1a'. The 'Format type' section has 'Formatted' selected, and the 'Pivot type' section has 'Header' selected. A 'Run' button is visible. The main area displays the 'Table name' as 'Table 1.1a Response rates (after substitution, unweighted)' and a 'Pivot table preview' of a table with 12 columns and 12 rows of household data.

	Area			Region					Degree of urbanization		
	Total	Urban	Rural	Western	Khangai	Central	Eastern	Ulaanbaatar	City	Town & Semi-dense area	Rural area
(Households sampled) Households sampled	3,000	1,557	1,443	545	547	594	595	719	623	930	1,447
(Households sampled) Interviewed	2,681	1,387	1,294	499	498	529	522	633	539	838	1,304
(Households sampled) No consent for next wave	97	68	29	10	14	18	17	38	37	35	25
(Households sampled) With consent for next wave	2,584	1,319	1,265	489	484	511	505	595	502	803	1,279
(Households sampled) Refused	73	52	21	4	9	14	14	32	28	24	21
(Households sampled) No consent for next wave	63	45	18	3	9	12	13	26	24	21	18
(Households sampled) With consent for next wave	10	7	3	1	0	2	1	6	4	3	3
(Households sampled) No eligible respondent	2	1	1	1	0	1	0	0	0	1	1
(Households sampled) Phone number(s) does not belong to sampled household	33	21	12	4	5	6	6	12	13	7	13
(Households sampled) Phone number(s) inactive	43	19	24	7	3	11	14	8	5	13	25

Figure 2: Tabulator: select a sheet on the left and click Run; the calculated table appears on the right.

1. Click **Tabulator** at the top.
2. Under **Select a sheet**, click the arrow. Choose the table code requested by your survey lead. The picture uses **1.1a**.
3. Leave **Format type** on **Formatted** and **Pivot type** on **Header** for an easy-to-read preview.
4. Click **Run** once and wait for the table to appear.
5. Read **Table name** above the result. Check that it is the table you intended to produce. Scroll inside the preview to see more rows or columns.

Ready to continue: the correct table name and calculated values appear. The next step is **Consistency Checks**. A preview has not yet saved your calculated table to an Excel file.

If your reviewer asks for plain numbers, choose **Unformatted**. Choose **Suppressed** to inspect the small-sample display rules. Keep **Header** selected unless your reviewer asks for another view.

5. Check the table

The screenshot shows the MICS Plus Tabulation web application. The top navigation bar includes links like 'User's Guide', 'Data Preparation', 'Tabulator', 'Consistency Checks', 'Write to Excel', 'Multi-Sheet Tabulator', 'Data Exploration', 'Data View', and 'Long Format Data'. The 'Consistency Checks' tab is active. On the left sidebar, there is a checkbox for 'Show only inconsistent cells' and a 'Run Checks' button. The main content area shows a 'Table name' field with 'Table 1.1a Response rates (after substitution, unweighted)'. Below this, there are several check tabs: 'Row Group (N)', 'Row Group (Perc)', 'Row Indent Group (N)', 'Row Indent Group (Perc)', 'Col Group (N)', and 'Col Group (Perc)'. The 'Row Group (N)' tab is selected. A table of results is displayed with the following columns: var_name_row, cond, col_index, stat_type, group_total, total_value, and diff_value. The table contains 11 rows for HH17 with col_index values from 3 to 15, all with diff_value of 0.

var_name_row	cond	col_index	stat_type	group_total	total_value	diff_value
HH17	1	3	n_unw	3000	3000	0
HH17	1	4	n_unw	1557	1557	0
HH17	1	5	n_unw	1443	1443	0
HH17	1	7	n_unw	545	545	0
HH17	1	8	n_unw	547	547	0
HH17	1	9	n_unw	594	594	0
HH17	1	10	n_unw	595	595	0
HH17	1	11	n_unw	719	719	0
HH17	1	13	n_unw	623	623	0
HH17	1	14	n_unw	930	930	0
HH17	1	15	n_unw	1447	1447	0

Figure 3: Consistency Checks: click Run Checks and examine the smaller check tabs and their result tables.

1. Click **Consistency Checks**. Confirm the **Table name** matches the table you just ran.
2. Click **Run Checks** and wait for the results.
3. Click each of the smaller check tabs, such as **Row Group (N)**. They compare counts or percentages in different parts of the table.
4. If a tab is red, tick **Show only inconsistent cells** to find the differences. Record the table code and ask your survey lead to review them.

Green: this check ran and found no difference beyond its tolerance. **Red:** a difference needs review. **No color or no rows:** the check may not apply; it does not mean it passed. Read the result as well as the color. The example's **diff_value** column shows zero where the compared totals agree.

Ready to continue: the applicable checks pass, or the survey lead has reviewed and accepted any exceptions. Passing checks do not replace the survey lead's review of the table and its meaning.

6. Save the reviewed table to Excel

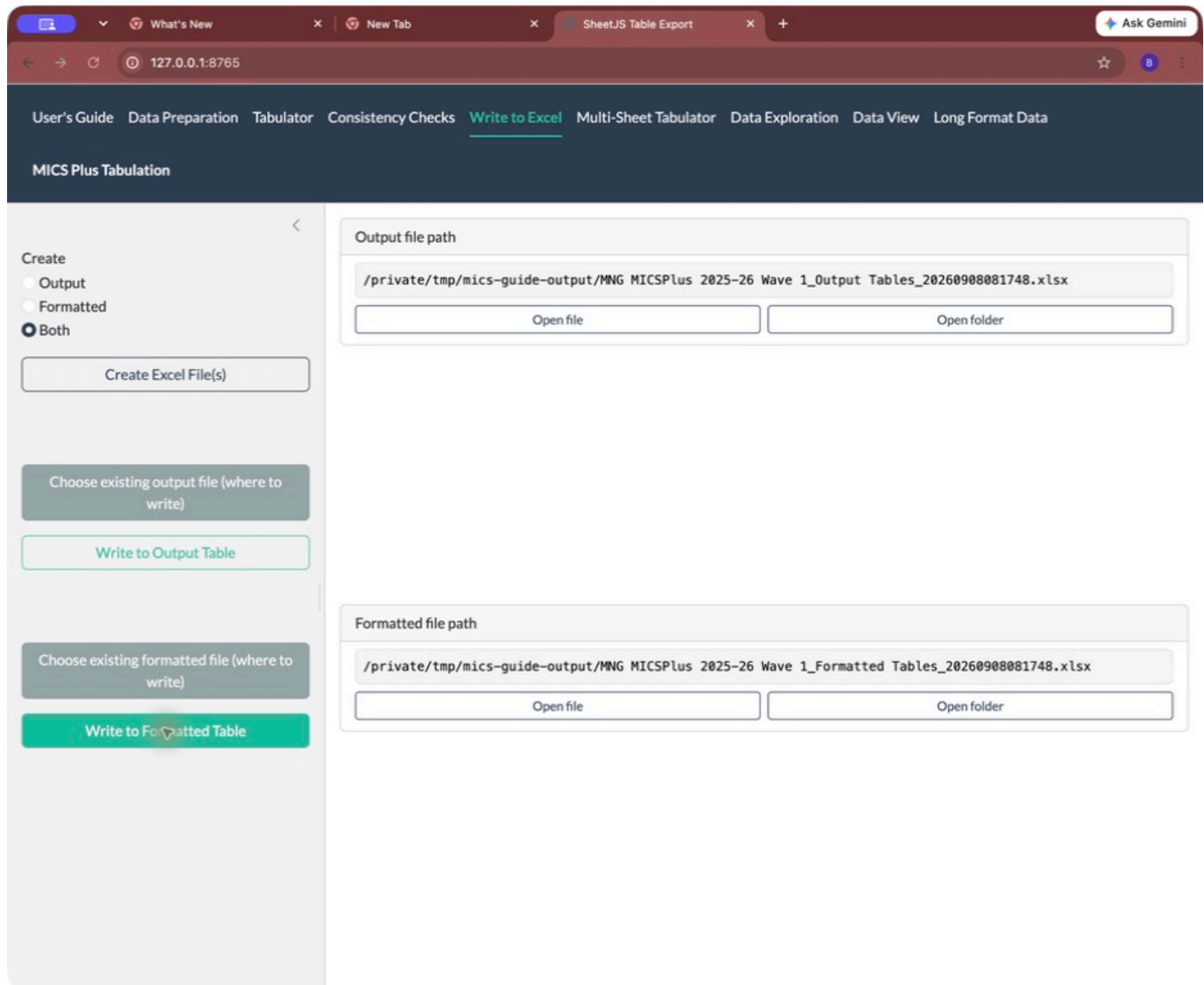


Figure 4: Write to Excel: create Both workbooks, write the current table with the two write buttons, and find the saved files using Open folder.

1. Click **Write to Excel**. Under **Create**, choose **Both** unless your survey lead requests one type only.
2. Click **Create Excel File(s)** once for this run. Two file paths appear on the right. This creates workbook copies; it does not fill every table.
3. Click **Write to Output Table**. Wait for the write to finish.
4. Click **Write to Formatted Table**. Wait again. Both buttons write the **current table** to its matching sheet.
5. Under **Formatted file path**, click **Open folder** to find the saved file, or **Open file** to view it. Check the expected sheet in Excel.

Output keeps numeric values for review. **Formatted** contains the table's presentation values and applicable notes. Close the workbook in Excel before writing another table. If **Open folder** does not work, copy the displayed folder path into File Explorer or Finder.

Next table: repeat **Tabulator > Run, Consistency Checks > Run Checks**, then the two **Write** buttons. Reuse the same workbooks; do not click **Create Excel File(s)** again for every table.

7. Run several tables together (optional)

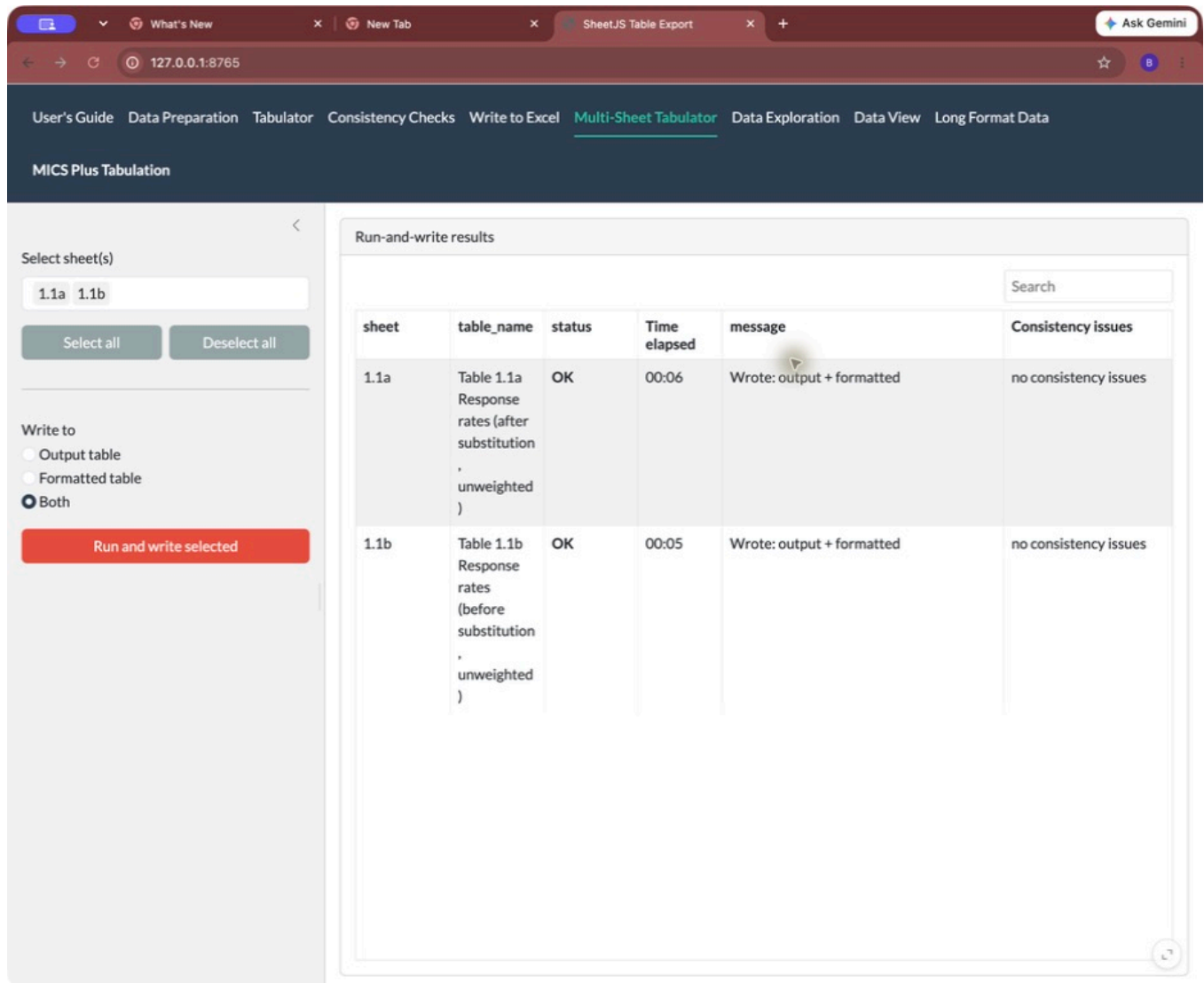


Figure 5: Multi-Sheet Tabulator: choose sheets, choose Both, then run and inspect one result row per sheet.

Use this after you can run and review one table successfully.

1. First create your workbook files in **Write to Excel** (step 6), if you have not already done so for this run.
2. Click **Multi-Sheet Tabulator**. Click **Select sheet(s)** and choose a table. Click again to add another. Use **Deselect all** to clear the selection.
3. Under **Write to**, choose **Both** or your agreed destination.
4. Click **Run and write selected** once. Keep RStudio and the browser open until it finishes.
5. When the completion message appears, click **OK**. Read every row in **Run-and-write results**, especially **status**, **message**, and **Consistency issues**. Use the horizontal scrollbar if a column is off the right edge.

Ready to continue: every requested table has completed and its results have been reviewed. This action writes tables as it runs; it does not wait for your review of each table. A completed write is not approval of a flagged result.

If a table fails or has consistency issues, note its code and message. Review it separately with your survey lead before sharing the workbook. Start with a small selection; use **Select all** only when you are ready for the full run.

8. Download a data file (optional)

The screenshot shows the 'Long Format Data' section of the MICS Plus Tabulation application. On the left, there is a sidebar with a 'Run all tables and prepare...' button and a 'Download output' section. The 'Download output' section includes a description 'Download all rows and columns of the prepared data.' and three buttons: 'Excel (.xlsx)', 'CSV (.csv)', and 'RDS (.rds)'. The main area displays an 'Output Table' with the following columns: STRUCTURE, STRUCTURE_ID, ACTION, FREQ, REF_AREA, SURVEY_ID, and TABL. The table contains 7 rows of data, all with 'dataflow' in the first column and 'MNG' in the fifth. The bottom of the table shows '1-100 of 18162 rows' and a 'Show 100' dropdown.

Figure 6: Long Format Data: prepare the data with the button on the left, then use a download button below it.

Use this only if your survey lead asks for the tables as a combined data file. For ordinary presentation tables, use **Write to Excel** instead.

1. Complete **Data Preparation** first.
2. Click **Long Format Data**.
3. Click **Run all tables and prepare....** Wait while the application calculates the tables. This may take longer than running one table.
4. When the data preview appears without an error, find **Download output** in the left panel. Click **Excel (.xlsx)** for a spreadsheet, or the format your recipient requested.
5. Find the file in your browser's **Downloads** list or your computer's **Downloads** folder. Some browsers ask you to choose a save location.

What you receive: one combined file of calculated table cells. It is a different layout from the **Formatted** workbook. **CSV (.csv)** is a plain text table; **RDS (.rds)** is for someone working in R.

The download contains all prepared rows, including rows hidden by preview filters or pages. Check the country, period, and wave in the filename before sending it. If preparation reports a failed table, ask your survey lead to resolve the error and rerun before relying on a download.

9. Look at prepared data (optional)

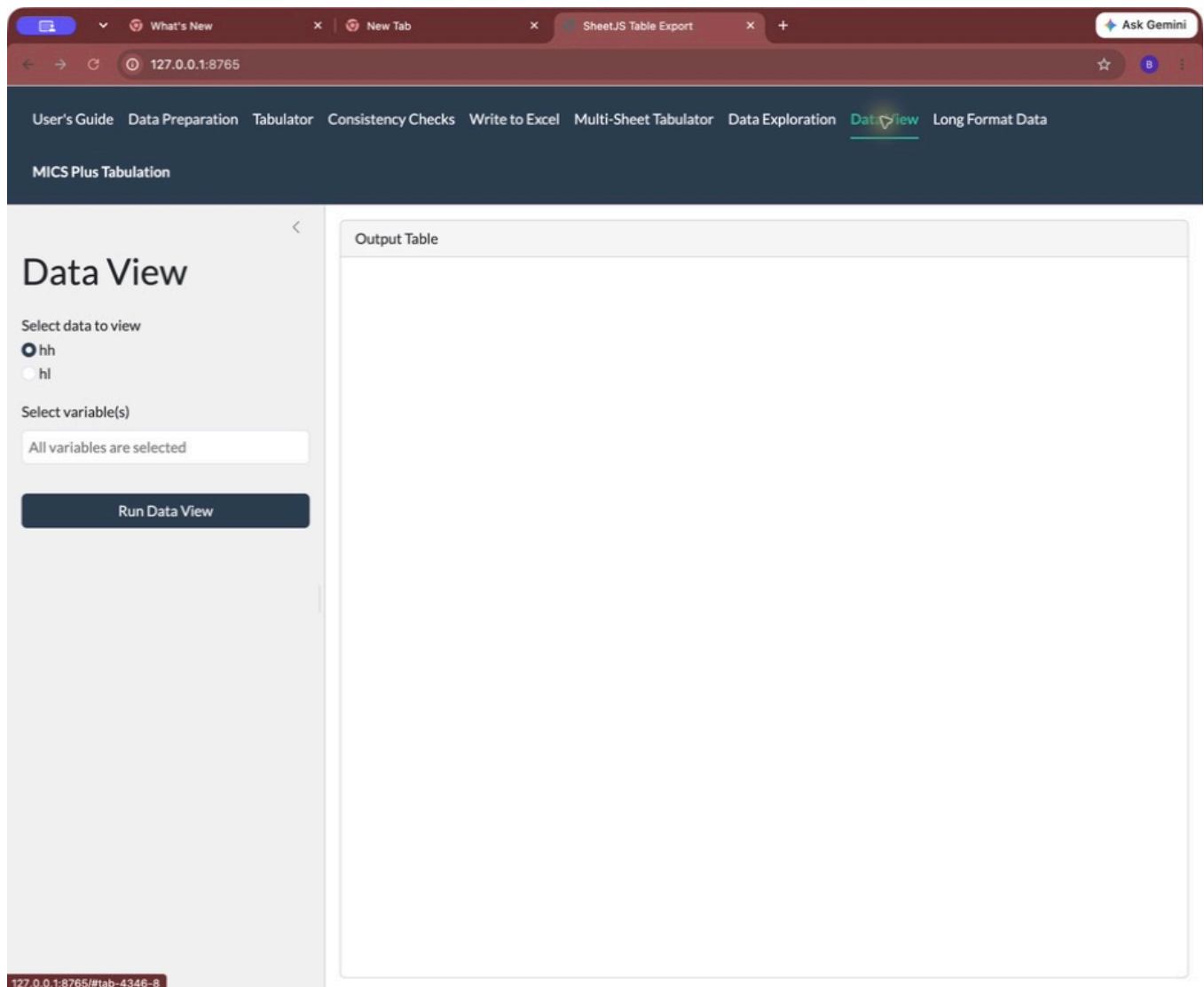


Figure 7: Data View: select the household or household members data, select variables, then click Run Data View.

This tool shows the data after preparation. Use it when your survey lead asks you to check a particular item. A **variable** is a named item in the data, like age or region.

1. Complete **Data Preparation**, then click **Data View**.
2. Under **Select data to view**, choose **hh** for households or **hl** for household members, as directed by your reviewer.
3. Select the variable names your reviewer gives you.
4. Click **Run Data View**. Read the preview and its labels.

Check: you are looking at the intended dataset and variables. This screen may show individual survey records. Share only what your survey team's data handling rules allow, and avoid including records in a help screenshot.

You can return to **Tabulator** by clicking its tab at the top. Viewing data here does not save a presentation table to Excel. The picture shows the controls before any individual records are displayed.

10. Explore a question (optional)

Figure 8: Data Exploration: choose the dataset, variables, and analysis type before clicking Run Data Explorer.

Use this for a question your survey lead wants to investigate. It does not replace the approved tabulation plan.

1. After **Data Preparation**, click **Data Exploration**.
2. Under **Select data set**, choose **hh** for households or **hl** for household members. Under **Select variable(s)**, choose the names your reviewer gives you.
3. Choose **Unweighted** or **Weighted**, as your reviewer specifies. If weighted, enter their exact variable name in **Type weight variable**. Do not guess it.
4. Choose **Frequency** to count categories of one variable, or **Cross table** to compare two variables. Select the second variable if requested.
5. Leave any filter box blank unless your reviewer supplies the exact text to use. Click **Run Data Explorer** and read the result.

Check: the selected data, variables, and weighting match the question you were asked to explore. These choices can change the meaning of a result. Ask your reviewer before treating it as a final survey finding.

11. Finish, return later, or get help

Finish your work

1. Write every reviewed table you want to keep to Excel (step 6).
2. Find the saved workbook. Open it and check the country, wave, and expected sheets. Keep your review notes with it.
3. Close the workbook, then close the application's browser tab.
4. In RStudio, click the red **Stop** button in the Console to stop the app. You may then close RStudio. Saving an R workspace is not a substitute for saving the Excel workbooks.

When you return: start the app again and select your inputs again. The app does not automatically restore the previous browser session. For a fresh run, create new workbook files. To continue an existing workbook, ask your survey lead to help select it using the **Choose existing...** buttons in **Write to Excel**. Writing the same sheet again replaces its previously written values.

If something goes wrong

What you see	What to do next
Missing file or wrong filename	Return to Data Preparation , select the correct file, and click Set again. Rerun and recheck the table.
Country or wave mismatch	Compare your dropdown choices with the files supplied by your team. Ask for a matched set if unsure.
Preparation error or missing variable	Record the message and send it to your survey lead. Do not edit the preparation code yourself.
Empty preview or no table	Check that Set completed, choose a sheet, and click Run . If the workbook preview alone is blank, check your connection and report that detail.
Red consistency tab	Record the table code and the displayed differences. Have your survey lead review them before sharing.
Excel file cannot be written	Close that workbook in Excel, then try its Write button again.
Cannot find the saved workbook	Use Write to Excel > Open folder or the path shown there. For a data download, check Downloads instead.
Browser disconnected or refreshed	Save any work still available. Restart the app if needed and select the inputs again; check existing files before rewriting.

When asking for help, include: country and wave, table code, the button you clicked, and the exact error text. A screenshot of the message helps; leave individual survey records out of it. Your support person can check the installed package version in RStudio.